1. What are Pseudo Classes?

Pseudo classes are keywords added to selectors using **single colon** `:`.

They define a **special state** of an element.

Syntax:

```
selector:pseudo-class {
   property: value;
}
```

Example:

```
button:hover {
   background: blue;
}
```

> Means: When mouse comes over button, make background blue.

Real Life Meaning

Think of button:

- Normal button
- Mouse on button
- Clicking button
- Visited link
- First child in list

These are **states**.

Pseudo class styles these states.

Common Pseudo Classes

---

1. `:hover`

When mouse comes over element.

```
button:hover {
   color: red;
}
```

```
<button>Click</button>
```

---

2. `:active`

When element is being clicked.

```
button:active {
   transform: scale(0.9);
}
```

---

3. `:focus`

When input gets selected.

```
input:focus {
   border: 2px solid blue;
}
```

Useful for forms.

---

4. `:visited`

For visited links.

```
a:visited {
   color: purple;
}
```

---

5. `:link`

Unvisited links.

```
a:link {
   color: green;
}
```

---

6. `:first-child`

Select first child.

```
li:first-child {
   color: red;
}
```

```
<ul>
  <li>One</li>
  <li>Two</li>
</ul>
```

Only "One" becomes red.

---

7. `:last-child`

```
li:last-child {
```

```
   color: blue;
}
```

---

8. `:nth-child(n)`

Select specific child.

```
li:nth-child(2) {
   color: orange;
}
```

Second li selected.

Even / Odd

```
tr:nth-child(even)
tr:nth-child(odd)
```

Used in tables.

---

9. `:not()`

Select except something.

```
p:not(.special) {
   color: gray;
}
```

All p except special class.

---

10. `:checked`

For checkbox/radio checked state.

```
input:checked {
   width: 30px;
}
```

---

11. `:disabled`

```
input:disabled {
   background: gray;
}
```

---

12. `:required`

```
input:required {
   border: 1px solid red;
}
```

What are Pseudo Elements?

Pseudo elements style **part of element**.

Use **double colon** `::`

```
selector::pseudo-element {
}
```

Why Double Colon?

To differentiate from pseudo classes.

Old CSS allowed single colon also:

```
:before
```

Modern CSS:

```
::before
```

Common Pseudo Elements

---

1. `::before`

Insert content before element.

```
p::before {
   content: "   ";
}
```

```
<p>Hello</p>
```

Output:

Hello

---

2. `::after`

Insert content after.

```
p::after {
   content: " ✓";
}
```

Output:

Hello ✓

Important Rule:

`content` required in before/after.

```
content: "";
```

Even empty content allowed.

---

3. `::first-letter`

Style first letter.

```
p::first-letter {
   font-size: 40px;
   color: red;
}
```

---

4. `::first-line`

Style first line.

```
p::first-line {
   font-weight: bold;
}
```

---

5. `::selection`

When user selects text.

```
::selection {
   background: yellow;
   color: black;
}
```

---

6. `::placeholder`

Style placeholder text.

```
input::placeholder {
   color: gray;
}
```

Difference Between Both

Pseudo Class	Pseudo Element
State of element	Part of element
Uses `:`	Uses `::`
Example `:hover`	Example `::before`

Easy Memory Trick

Pseudo Class = Behavior / Condition

- Hover
- Focus
- Checked
- First child

Pseudo Element = Virtual Part

- Before
- After
- First letter
- Selection

Combined Example

```
<button>Submit</button>
```

```
button:hover::after {
   content: "   ";
}
```

When hover button:

Submit

Deep Real Projects Usage

Button Hover Effect

```
button {
   background: black;
   color: white;
}

button:hover {
   background: red;
}
```

Input Focus

```
input:focus {
   outline: none;
   border-color: blue;
}
```

Fancy Underline using `::after`

```
a {
   position: relative;
}

a::after {
   content: "";
   position: absolute;
   width: 0;
   height: 2px;
   background: red;
   left: 0;
   bottom: 0;
   transition: 0.3s;
}

a:hover::after {
   width: 100%;
}
```

Used in navbar animations.

Child Selection Example

```
li:nth-child(odd) {
   background: lightgray;
}
```

Interview Questions

Q1 Why pseudo used?

To style without extra HTML or JS.

Q2 Why `::before` useful?

Can add icons, lines, design effects.

Q3 Difference between `:nth-child()` and `:nth-of-type()`?

- `nth-child()` counts all children
- `nth-of-type()` counts same type only

Important Mistakes Beginners Make

Wrong:

```
div::hover
```

Correct:

```
div:hover
```

Wrong:

```
div:before
```

Allowed old syntax but better:

```
div::before
```

Full Example

```
<ul>
  <li>One</li>
  <li>Two</li>
  <li>Three</li>
</ul>
```

```
li:first-child {
 color:red;
}

li:last-child {
 color:blue;
}

li:nth-child(2)::after {
 content:" ⋆";
}
```

Output:

One red
Two ⋆
Three blue

Shortcut Revision

Pseudo Classes

```
:hover
:focus
:active
:first-child
:last-child
:nth-child()
:not()
:checked
```

Pseudo Elements

```
::before
::after
::first-letter
::first-line
::selection
::placeholder
```

Best Way to Practice

Create:

1. Hover button
2. Animated navbar underline
3. Zebra table rows
4. Fancy paragraph first letter
5. Custom checkbox

Final Summary

Pseudo classes = **state change**

Pseudo elements = **virtual internal parts**

Both are heavily used in frontend interviews and UI design.